

Lab 06: Black Box Test Plan

Test Overview

To run the tests:

1. Right click on PackSchedulerGUI class in the Package Explorer.
2. Select Run As > Java Application

All test files should be located in the test-files/ folder.

The registrar's information is provided in the registrar.properties file.

If you run all tests independently, you can close the GUI between each test. However, the tests are structured so that they may be run in order without shutting down the GUI.

Test ID	Description	Expected Results	Actual Results
T1: Invalid Login	Preconditions: PackSchedulerGUI has loaded User ID: Iberg Password: pw Click Login Click OK Click Clear	Pop up with message "User doesn't exist." User ID and password text fields are cleared.	Pop up with message "User doesn't exist." User ID and password text fields are cleared when user clicks clear.
T2: Registrar	Preconditions: PackSchedulerGUI has loaded		

Login	User ID: name as provided in registrar.properties Password: password as provided in registrar.properties Click Login	View switches to Registrar panel with Student Directory functionality displayed	View switches to Registrar panel with Student Directory functionality displayed
T3: Load Student Directory	Preconditions: PackSchedulerGUI has loaded and registrar user is logged in (T2) Click Load Student Directory Select test-files/student_records.txt Click Select	10 students are listed in alphabetical order by last name. The list starts with Demetrius Austin and ends with Griffith Stone.	10 students are listed in alphabetical order by last name. The list starts with Demetrius Austin and ends with Griffith Stone.
T4: Add Invalid Student - no first name	Preconditions: PackSchedulerGUI has loaded and registrar user is logged in (T2). T3 has run. Enter the following in the Student Information section: • First Name: • Last Name: Wilkins • ID: pwilkins • Email: pwilkins@ncsu.edu • Password: pw • Repeat Password: pw • Max Credits: 15 Click Add Student Click OK	Error Message: Invalid first name	Error Message: Invalid first name

T5: Add Invalid Student - no last name	Preconditions: PackSchedulerGUI has loaded and registrar user is logged in (T2). T3 has run. Enter the following in the Student Information section: • First Name: Patience • Last Name: • ID: pwilkins • Email: pwilkins@ncsu.edu • Password: pw • Repeat Password: pw • Max Credits: 15 Click Add Student Click OK	Error Message: Invalid last name	Error Message: Invalid last name
T6: Add Invalid Student - no id	Preconditions: PackSchedulerGUI has loaded and registrar user is logged in (T2). T3 has run. Enter the following in the Student Information section: • First Name: Patience • Last Name: Wilkins • ID: • Email: pwilkins@ncsu.edu • Password: pw • Repeat Password: pw • Max Credits: 15 Click Add Student Click OK	Error Message: Invalid id	Error Message: Invalid id
T7: Add Invalid	Preconditions: PackSchedulerGUI has loaded and registrar user is logged in (T2). T3 has run.		

Student - no email	Enter the following in the Student Information section: • First Name: Patience • Last Name: Wilkins • ID: pwilkins • Email: • Password: pw • Repeat Password: pw • Max Credits: 15 Click Add Student Click OK	Error Message: Invalid email	Error Message: Invalid email
T8: Add Invalid Student - invalid email	PPreconditions: PackSchedulerGUI has loaded and registrar user is logged in (T2). T3 has run. Enter the following in the Student Information section: • First Name: Patience • Last Name: Wilkins • ID: pwilkins • Email: pwilkinsncsu.edu • Password: pw • Repeat Password: pw • Max Credits: 15 Click Add Student Click OK	Error Message: Invalid email	Error Message: Invalid email
T9: Add Invalid Student - password	Preconditions: PackSchedulerGUI has loaded and registrar user is logged in (T2). T3 has run. Enter the following in the Student Information		

s don't match	section: • First Name: Patience • Last Name: Wilkins • ID: pwilkins • Email: pwilkins@ncsu.edu • Password: pw1 • Repeat Password: pw • Max Credits: 15 Click Add Student Click OK	Error Message: Passwords do not match	Error Message: Passwords do not match
T10: Add Invalid Student - zero max credits	Preconditions: PackSchedulerGUI has loaded and registrar user is logged in (T2). T3 has run. Enter the following in the Student Information section: • First Name: Patience • Last Name: Wilkins • ID: pwilkins • Email: pwilkins@ncsu.edu • Password: pw • Repeat Password: pw • Max Credits: 0 Click Add Student Click OK	Patience Wilkins is added to the Student Directory and is the last student in the list. Max credits is set to 18.	Patience Wilkins is added to the Student Directory and is the last student in the list. Max credits is set to 18.
T11: Add Invalid Student - 19 max	Preconditions: PackSchedulerGUI has loaded and registrar user is logged in (T2). T3 has run. Enter the following in the Student Information		

credits	section: • First Name: Alice • Last Name: Adams • ID: aadams • Email: aadams@ncsu.edu • Password: pw • Repeat Password: pw • Max Credits: 19 Click Add Student Click OK	Alice Adams is added to the Student Directory and is the first student in the list. Max credits is set to 18.	Alice Adams is added to the Student Directory and is the first student in the list. Max credits is set to 18.
T12: Add Student - end	Preconditions: PackSchedulerGUI has loaded and registrar user is logged in (T2). T3 has run. Enter the following in the Student Information section: • First Name: Mikayla • Last Name: William • ID: mwilliam • Email: mwilliam@ncsu.edu • Password: pw • Repeat Password: pw • Max Credits: 15 Click Add Student Click OK	Mikayla William is added to the Student Directory and is the last student in the list.	Mikayla William is added to the Student Directory and is the last student in the list.
T13: Add Student - middle	Preconditions: PackSchedulerGUI has loaded and registrar user is logged in (T2). T3 has run.		

	Enter the following in the Student Information section: • First Name: Rylee • Last Name: Puckett • ID: rpuckett • Email: rpuckett@ncsu.edu • Password: pw • Repeat Password: pw • Max Credits: 15 Click Add Student Click OK	Rylee Puckett is added to the Student Directory between Dylan Nolan and Cassandra Schwartz	Rylee Puckett is added to the Student Directory between Dylan Nolan and Cassandra Schwartz
T14: Add Student - front	Preconditions: PackSchedulerGUI has loaded and registrar user is logged in (T2). T3 has run. Enter the following in the Student Information section: • First Name: Melvin • Last Name: Acevedo • ID: macevedo • Email: macevedo@ncsu.edu • Password: pw • Repeat Password: pw • Max Credits: 15 Click Add Student Click OK	Melvin Acevedo is added to the Student Directory and is the first student in the list.	Melvin Acevedo is added to the Student Directory and is the first student in the list.
T15: Remove	Preconditions: PackSchedulerGUI has loaded and registrar user is logged in (T2).		

Student - no selection	Click Remove Student	Error Message: No student selected	Error Message: No student selected
T16: Remove Student - middle	Preconditions: PackSchedulerGUI has loaded and registrar user is logged in (T2). Select Shannon Hansen Click Remove student	Shannon Hansen is removed. Athea Hicks follows Emerald Frost in the list	Shannon Hansen is removed. Athea Hicks follows Emerald Frost in the list
T17: Remove Student - front	Preconditions: PackSchedulerGUI has loaded and registrar user is logged in (T2). T14 has run. Select Melvin Acevedo Click Remove student	Melvin Acevedo is removed. Demetrius Austin is front of the directory	Melvin Acevedo is removed. Alice Adams is front of the directory
T18: Remove Student - back	Preconditions: PackSchedulerGUI has loaded and registrar user is logged in (T2). T12 has run. Select Mikayla William Click Remove student	Mikayla William is removed. Griffith Stone is last in the directory	Mikayla William is removed. Patience Wilkins is last in the directory
T19:	Preconditions: PackSchedulerGUI has loaded and	Contents should be (note that	

Save Student Directory	registrar user is logged in (T2). T12-T18 have run. Click Save Student Directory Select test-files directory and name the file t19_student_directory.txt.	the password hash may be different due to encoding): Alice,Adams,aadams,aadams@ncsu.edu,0ÉRú±"ÄùuŸ!Û\7X²F'pâ9•{-OîFâapÄ,18Demetrius,Austin,daustin,Curabitur.egestas.nunc@placatorcilacus.co.uk,0ÉRú±"ÄùuŸ!Û\7X²F'pâ9•{-OîFâapÄ,18Lane,Berg,lberg,sociis@non.org,0ÉRú±"ÄùuŸ!Û\7X²F'pâ9•{-OîFâapÄ,14Raymond,Brennan,rbrennan,lit ora.torquent@pellentesquema ssalobortis.ca,0ÉRú±"ÄùuŸ!Û\7X²F'pâ9•{-OîFâapÄ,12Emerald,Frost,efrost,adipiscing@acipsumPhasellus.edu,0ÉRú±"ÄùuŸ!Û\7X²F'pâ9•{-OîFâapÄ,3Althea,Hicks,ahicks,Phasellus.dapibus@luctusfelis.com,0ÉRú±"ÄùuŸ!Û\7X²F'pâ9•{-OîFâapÄ,11Zahir,King,zking,orci.Donec@ametmassaQuisque.com,0ÉRú±"ÄùuŸ!Û\7X²F'pâ9•{-OîFâapÄ,15Dylan,Nolan,dnolan,placerat.Cras.dictum@dictum.net,0ÉRú±"ÄùuŸ!Û\7X²F'pâ9•{-OîFâapÄ,5	Alice,Adams,aadams,aadams@ncsu.edu,MMIS+rEiw/l1nwKm2Vw3WLJGtP7iOZV7LU/uRuJhcMQ=,18Demetrius,Austin,daustin,Curabitur.egestas.nunc@placatorcilacus.co.uk,MMIS+rEiw/l1nwKm2Vw3WLJGtP7iOZV7LU/uRuJhcMQ=,18Lane,Berg,lberg,sociis@non.org,M MIS+rEiw/l1nwKm2Vw3WLJGtP7iOZV7LU/uRuJhcMQ=,14Raymond,Brennan,rbrennan,lit ora.torquent@pellentesquemasalobortis.ca,MMIS+rEiw/l1nwKm2Vw3WLJGtP7iOZV7LU/uRuJhcMQ=,12Emerald,Frost,efrost,adipiscing@acipsumPhasellus.edu,MMIS+rEiw/l1nwKm2Vw3WLJGtP7iOZV7LU/uRuJhcMQ=,3Althea,Hicks,ahicks,Phasellus.dapibus@luctusfelis.com,MMIS+rEiw/l1nwKm2Vw3WLJGtP7iOZV7LU/uRuJhcMQ=,11Zahir,King,zking,orci.Donec@ametmassaQuisque.com,M MIS+rEiw/l1nwKm2Vw3WLJGtP7iOZV7LU/uRuJhcMQ=,15Dylan,Nolan,dnolan,placerat.Cras.dictum@dictum.net,MMIS+rEiw/l1nwKm2Vw3WLJGtP7iOZV7LU/uRuJhcMQ=,5Rylee,Puckett,rpuckett,rpuckett@ncsu.edu,MMIS+rEiw/l1nwKm2Vw3W
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		Rylee,Puckett,rpuckett,rpuckett@ncsu.edu,0ÉRú±"ÃùuŸ!Û\7X²F'pâ9•{-OîFâapÄ,15 Cassandra,Schwartz,cschwartz,semper@imperdietornare.co.uk,0ÉRú±"ÃùuŸ!Û\7X²F'pâ9•{-OîFâapÄ,4 Griffith,Stone,gstone,porta@magnamalesuadavel.net,0ÉRú±"ÃùuŸ!Û\7X²F'pâ9•{-OîFâapÄ,17 Patience,Wilkins,pwilkins,pwilkins@ncsu.edu,0ÉRú±"ÃùuŸ!Û\7X²F'pâ9•{-OîFâapÄ,18	LJGtP7iOZV7LU/uRuJhcMQ=,15Cassandra,Schwartz,cschwartz,semper@imperdietornare.co.uk,MMIS+rEiw/l1nwKm2Vw3WLJGtP7iOZV7LU/uRuJhcMQ=,4Griffith,Stone,gstone,porta@magnamalesuadavel.net,MMIS+rEiw/l1nwKm2Vw3WLJGtP7iOZV7LU/uRuJhcMQ=,17Patience,Wilkins,pwilkins,pwilkins@ncsu.edu,MMIS+rEiw/l1nwKm2Vw3WLJGtP7iOZV7LU/uRuJhcMQ=,18
T20: New Student Directory	Preconditions: PackSchedulerGUI has loaded and registrar user is logged in (T2). T12-T18 have run. Click New Student Directory	An empty student directory is shown.	An empty student directory is shown.
T21: Add Student - first	Preconditions: PackSchedulerGUI has loaded and registrar user is logged in (T2). T20 has run. Enter the following in the Student Information section: • First Name: Patience • Last Name: Wilkins • ID: pwilkins • Email: pwilkins@ncsu.edu • Password: pw • Repeat Password: pw • Max Credits: 15 Click Add Student	Patience Wilkins is added to	Patience Wilkins is added to

	Click OK	the Student Directory and is the only student in the list.	the Student Directory and is the only student in the list.
T22: Course Catalog view	Preconditions: PackSchedulerGUI has loaded and registrar user is logged in (T2). Click Course Catalog	Course Catalog functionality displayed	
T23: Load Course Catalog	Preconditions: PackSchedulerGUI has loaded and registrar user is logged in (T2) Click Load Course Catalog Select test-files/course_records.txt Click Select	13 courses are listed starting with CSC116-001 and ending with CSC316-001	13 courses are listed starting with CSC116-001 and ending with CSC316-001
T24: Add Invalid Course - no name	Preconditions: PackSchedulerGUI has loaded and registrar user is logged in (T2). T23 has run. Enter the following in the Course Information section: • Course Name: • Course Title: Computer Organization and Assembly Language • Course Section: 001 • Credits: 3 • Instructor Id: lasher • Start Time: 10:15 AM • End Time: 11:30 AM • Meeting Days: Tuesday & Thursday Click Add Course	Error Message: Invalid course name	Error Message: Invalid course name

	Click OK		
T25: Add Invalid Course - no title	Preconditions: PackSchedulerGUI has loaded and registrar user is logged in (T2). T23 has run. Enter the following in the Course Information section: • Course Name: CSC236 • Course Title: • Course Section: 001 • Credits: 3 • Instructor Id: lasher • Start Time: 10:15 AM • End Time: 11:30 AM • Meeting Days: Tuesday & Thursday Click Add Course Click OK	Error Message: Invalid title.	Error Message: Invalid course name
T26: Add Invalid Course - no section	Preconditions: PackSchedulerGUI has loaded and registrar user is logged in (T2). T23 has run. Enter the following in the Course Information section: • Course Name: CSC236 • Course Title: Computer Organization and Assembly Language • Course Section: • Credits: 3 • Instructor Id: lasher • Start Time: 10:15 AM		

	• End Time: 11:30 AM • Meeting Days: Tuesday & Thursday Click Add Course Click OK	Error Message: Invalid section	Error Message: Invalid course name
T27: Add Invalid Course - section too short	Preconditions: PackSchedulerGUI has loaded and registrar user is logged in (T2). T23 has run. Enter the following in the Course Information section: • Course Name: CSC236 • Course Title: Computer Organization and Assembly Language • Course Section: 0011 • Credits: 3 • Instructor Id: lasher • Start Time: 10:15 AM • End Time: 11:30 AM • Meeting Days: Tuesday & Thursday Click Add Course Click OK	Error Message: Invalid section	Error Message: Invalid course name
T28: Add Invalid Course - section too long	Preconditions: PackSchedulerGUI has loaded and registrar user is logged in (T2). T23 has run. Enter the following in the Course Information section: • Course Name: CSC236		

	- Course Title: Computer Organization and Assembly Language - Course Section: 01 - Credits: 3 - Instructor Id: lasher - Start Time: 10:15 AM - End Time: 11:30 AM - Meeting Days: Tuesday & Thursday Click Add Course Click OK	Error Message: Invalid section	Error Message: Invalid course name
T29: Add Invalid Course - no instructor id	Preconditions: PackSchedulerGUI has loaded and registrar user is logged in (T2). T23 has run. Enter the following in the Course Information section: - Course Name: CSC236 - Course Title: Computer Organization and Assembly Language - Course Section: 001 - Credits: 3 - Instructor Id: - Start Time: 10:15 AM - End Time: 11:30 AM - Meeting Days: Tuesday & Thursday Click Add Course Click OK	Error Message: Invalid instructor id.	Error Message: Invalid course name
T30: Add Invalid	Preconditions: PackSchedulerGUI has loaded and registrar user is logged in (T2). T23 has run.		

Course - end time after start time	Enter the following in the Course Information section: • Course Name: CSC236 • Course Title: Computer Organization and Assembly Language • Course Section: 001 • Credits: 3 • Instructor Id: lasher • Start Time: 11:30 AM • End Time: 10:15 AM • Meeting Days: Tuesday & Thursday Click Add Course Click OK	Error Message: Invalid meeting days and times.	Error Message: Invalid meeting days and times.
T31: Add Invalid Course - no meeting days	Preconditions: PackSchedulerGUI has loaded and registrar user is logged in (T2). T23 has run. Enter the following in the Course Information section: • Course Name: CSC236 • Course Title: Computer Organization and Assembly Language • Course Section: 001 • Credits: 3 • Instructor Id: lasher • Start Time: 10:15 AM • End Time: 11:30 AM • Meeting Days: Click Add Course	Error Message: Invalid meeting days and times.	Error Message: Invalid course name

	Click OK		
T32: Add Course - end	Preconditions: PackSchedulerGUI has loaded and registrar user is logged in (T2). T23 has run. Enter the following in the Course Information section: • Course Name: CSC492 • Course Title: Senior Design • Course Section: 001 • Credits: 3 • Instructor Id: heil • Start Time: 9:30AM • End Time: 11:20AM • Meeting Days: Monday & Wednesday Click Add Course Click OK	CSC492-001 is added to the list after CSC316-001	Error Message: Invalid course name
T33: Add Course - front	Preconditions: PackSchedulerGUI has loaded and registrar user is logged in (T2). T23 has run. Enter the following in the Course Information section: • Course Name: CSC113 • Course Title: Introduction to Computing - MATLAB • Course Section: 001 • Credits: 3 • Instructor Id: tbdimitr • Start Time: 3:00 PM • End Time: 3:50 PM • Meeting Days: Monday & Wednesday		

	Click Add Course Click OK	CSC113-001 is added to the list at the front and is followed by CSC116-001.	Error Message: Invalid course name
T34: Add Course - middle	Preconditions: PackSchedulerGUI has loaded and registrar user is logged in (T2). T23 has run. Enter the following in the Course Information section: • Course Name: CSC116 • Course Title: Introduction to Programming - Java • Course Section: 006 • Credits: 3 • Instructor Id: jdyoung2 • Start Time: 10:40 AM • End Time: 12:30 PM • Meeting Days: Tuesday & Thursday Click Add Course Click OK	CSC116-006 is added to the list between CSC116-003 and CSC216-001.	Error Message: Invalid course name
T35: Remove Course - no selection	Preconditions: PackSchedulerGUI has loaded and registrar user is logged in (T2). Click Remove Course	Error Message: No course selected	Error Message: No course selected
T36: Remove Course - middle	Preconditions: PackSchedulerGUI has loaded and registrar user is logged in (T2). Select CSC216-001		

	Click Remove Course	CSC216-001 is removed. CSC116-006 is followed by CSC216-002.	CSC216-001 is removed. CSC116-006 is followed by CSC216-002.
T37: Remove Course - front	Preconditions: PackSchedulerGUI has loaded and registrar user is logged in (T2). T33 has run. Select CSC113-001 Click Remove Course	CSC113-001 is removed. CSC116-001 is the first course in the catalog	No CSC113-001 because couldnt add previously due to no course name error
T38: Remove Course - back	Preconditions: PackSchedulerGUI has loaded and registrar user is logged in (T2). T32 has run. Select CSC492-001 Click Remove Course	CSC492-001 is removed. CSC316-001 is the last course in the catalog	No CSC492-001 because couldn't add previously due to no course name error
T39: Save Course Catalog	Preconditions: PackSchedulerGUI has loaded and registrar user is logged in (T2). T32-T38 have run. Click Save Course Catalog Select test-files directory and name the file	Contents should be: CSC116,Intro to Programming - Java,001,3,jdyoung2,MW,910, 1100	No course catalof saved with t39_course_catalog.txt

	t39_course_catalog.txt.	CSC116,Intro to Programming - Java,002,3,spbalik,MW,1120,1310 CSC116,Intro to Programming - Java,003,3,tbdimitr,TH,1120,1310 CSC116,Introduction to Programming - Java,006,3,jdyoung2,TH,1040,1230 CSC216,Software Development Fundamentals,002,3,ixdoming,MW,1330,1445 CSC216,Software Development Fundamentals,601,3,jctetter,A CSC217,Software Development Fundamentals Lab,202,1,sesmith5,M,1040,1230 CSC217,Software Development Fundamentals Lab,211,1,sesmith5,T,830,1020 CSC217,Software Development Fundamentals Lab,223,1,sesmith5,W,1500,1650 CSC217,Software Development Fundamentals	
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		Lab,601,1,sesmith5,A CSC226,Discrete Mathematics for Computer Scientists,001,3,tmbarnes,MW F,935,1025 CSC230,C and Software Tools,001,3,dbsturgi,MW,1145, 1300 CSC316,Data Structures and Algorithms,001,3,jtking,MW,83 0,945	
T40: New Course Catalog	Preconditions: PackSchedulerGUI has loaded and registrar user is logged in (T2). T12-T18 have run. Click New Course Catalog	An empty course catalog is shown.	An empty course catalog is shown.
T41: Add Course - first	Preconditions: PackSchedulerGUI has loaded and registrar user is logged in (T2). T20 has run. Enter the following in the Course Information section: • Course Name: CSC216 • Course Title: Programming Concepts - Java • Course Section: 601 • Credits: 4 • Instructor Id: jctetter • Start Time: • End Time: • Meeting Days: Arranged Click Add Course	CSC216-601 is added to the	Error Message: Invalid Course Name

	Click OK	list	
T42: Logout	Preconditions: PackSchedulerGUI has loaded and registrar user is logged in (T2). Click Logout	User is returned to the Login view	User is returned to the Login view
T42: Login Student	Preconditions: PackSchedulerGUI has loaded and registrar user is logged in (T2). T21 and T42 have run. User ID: pwilkins Password: pw Click: Login	pwilkins is logged in and is switched to the user view. All that is shown is a Logout button.	pwilkins is logged in and is switched to the user view. All that is shown is a Logout button.